Table 4: Comparison with content-agnostic methods on CGL-V2. In this table, we compare Graphist based on Qwen1.5-7B [3] against other content-agnostic methods. Content-agnostic refers to generating a layout based solely on the background image. C-R means Composition-relevant.

Method	C-R measures		Graphic measures		
	$R_{com} \downarrow$	$R_{occ} \uparrow$	$R_{ali} \downarrow$	$R_{ove} \downarrow$	$R_{und} \uparrow$
BLT [43]	28.540	**1.000**	0.004	0.002	0.993
LayoutDM [35]	21.300	**1.000**	0.006	0.039	0.896
RADM [46]	10.260	0.997	0.008	0.046	0.983
Graphist*	**7.074**	**1.000**	**0.0003**	**0.001**	**1.000**

Table 5: Different LLM model. Comparison of Qwen1.5-0.5B, Qwen1.5-7B, and InternLM2-7B performance on the Crello dataset.

LLM	S_{DL}	S_{GI}	S_{IO}	S_{TV}	IOPR_{min}	IOPR_{avg}
Qwen1.5-0.5B	5.44	6.35	4.94	4.60	0.641	0.963
Qwen1.5-7B	5.85	6.90	5.10	5.37	0.667	0.971
InternLM2-7B	5.60	6.50	4.90	5.00	0.476	0.966

for automatic layout generation, exhibiting notable advancements in synthesizing information-rich graphic compositions.

5.4 Ablation Studies

In this section, we delve into a series of ablation studies to examine the underlying factors that influence the layout quality of Graphist. The specific aspects under investigation include the sequencing of input design elements, the choice of language model, the number of visual tokens, and the encoding of transparency through RGB-A image channels.

Influence of input layer sequencing. In Table 2, we show the performance comparison of our method with structured versus unstructured input sequences. The findings indicate that our approach achieves better results in the metrics S_{DL}, S_{GI}, and S_{IO} when dealing with an unordered input. The flexibility introduced by allowing randomness in input ordering contributes to the superior performance observed in S_{DL} and S_{IO} metrics versus a structured sequence. Nevertheless, this unordered approach increases the complexity of text layer placement, reflected by a reduced score in S_{TV}.

Fig. 6: More user-generated designs via graphist web demo. To evaluate usability, we invited numerous non-expert volunteers to upload their design components to our Graphist web demo, resulting in the creation of their own design projects. The image displays a selection of these high-quality outcomes alongside their respective layouts. Within the layout map, various colors signify distinct layer attributes as recognized by the model.

Impact of LLM. In this evaluation, we explore the influence of varying LLM configurations on our layout generation results. Similarly, we carry out experiments using the Crello dataset and the CGL-V2 dataset. We evaluate models with differing capacities: Qwen1.5-0.5B, Qwen1.5-7B, and InternLM2-7B. As demonstrated in Table 5 and Table 6, our experiments underline that within the domain of our tasks, the larger models consistently outperform their smaller counterparts. In addition, different models of the same size also reflect performance differences.

Influence of visual token length. In our method, we typically represent an image using a sequence of 5 visual tokens. To evaluate the effects of increased token quantity, potentially enhancing the image representation's granularity, we conducted experiments using a 17 tokens format. The composition of 17 tokens includes 1 cls token and 16 visual tokens, achieved by using a 4×4 pooling kernel size. The results presented in Table 7 suggests that lengthening the visual token sequence does not necessarily lead to performance improvements. According to the outcomes, the representation of an image with a quintet of tokens sufficiently captures the necessary information for this specific task.

Table 6: Different LLM model. Comparison of Qwen1.5-7B and InternLM2-7B performance on the CGL-V2 dataset.

LLM	Task	C-R measures		Graphic measures		
		$R_{com}\downarrow$	$R_{occ}\uparrow$	$R_{ali}\downarrow$	$R_{ove}\downarrow$	$R_{und}\uparrow$
InternLM2-7B	Aware	12.803	**1.000**	**0.004**	**0.002**	0.988
Qwen1.5-7B	Aware	12.150	**1.000**	**0.004**	**0.002**	**0.996**
InternLM2-7B	Agnostic	9.510	**1.000**	0.004	**0.001**	0.999
Qwen1.5-7B	Agnostic	**7.074**	**1.000**	**0.0003**	**0.001**	**1.000**

Table 7: Different visual token length. Comparison of different sequence length performance on the Crello dataset based on Graphist-Tiny. From our observation, five tokens are sufficient for layout generation.

Length	S_{DL}	S_{GI}	S_{IO}	S_{TV}	IOPR_{min}	IOPR_{avg}
5 tokens	5.44	6.35	4.94	4.60	0.641	0.963
17 tokens	5.47	6.28	4.88	4.67	0.667	0.965

RGB *vs*. RGB-A. To assess the effect of omitting the alpha channel, we have also tested the model with traditional three-channel RGB images. As illustrated in Table 8, inputs with the additional alpha channel (RGB-A) yield higher-quality outputs with more accurate layer ordering when compared to RGB inputs. The alpha channel provides detailed information that aids the model in discerning textural elements and gradients within the image layers. Importantly, when processing text layers, the alpha channel allows the model to isolate text from potentially distracting backgrounds, facilitating improved clarity and precision in text placement.

5.5 Real-World Evaluation

To assess the real-world applicability of our method, we engaged a group of non-expert volunteers to submit their design assets via our Graphist web demo and chose a selection of representative outputs for analysis. As illustrated in Figure 6, our model successfully generates visually appealing and cohesive designs across varying canvas dimensions and design elements.

5.6 Limitations & Negative Impact

In order to create an efficient design assistant that adheres to user intent, Graphist only represents a starting point. The automatic generation of complete sets of high-quality materials that meet design intent requirements, and